```
        System.out.println("\nEmployee Details:");
        System.out.println("----------------------------");

        for (String name : employees.keySet()) {
            System.out.println("Employee Name : " + name);
            System.out.println("Salary       : " + employees.get(name));
            System.out.println("----------------------------");
        }

        System.out.print("\nEnter Employee Name to Search: ");
        String searchName = sc.nextLine();

        if (employees.containsKey(searchName)) {
            System.out.println("Employee Name : " + searchName);
            System.out.println("Salary       : " + employees.get(searchName));
        } else {
            System.out.println("Employee not found.");
        }

        sc.close();
    }
}
```

Slip 4

1)Write a java program to check whether a given number is Positive or negative.

```
import java.util.Scanner;
```

```
public class PositiveNegative {
   public static void main(String args[]) {

      Scanner sc = new Scanner(System.in);

      System.out.print("Enter a number: ");
      int num = sc.nextInt();

      if (num > 0) {
         System.out.println("The number is Positive.");
      } else if (num < 0) {
         System.out.println("The number is Negative.");
      } else {
         System.out.println("The number is Zero.");
      }

      sc.close();
   }
}
```

2) Write a java program to define a class CricketPlayer(name, no_of_innings, no_of_times_notout, total_runs, bat_avg). Create an array of "n" player objects. Calculate the batting average for each player using a static method avg (). Handle appropriate exception while calculating average. Define static method "sortPlayer" which sort the array on the basis of average. Display the player details in sorted order.

```
import java.util.Scanner;

class CricketPlayer {
```

```
String name;
int no_of_innings;
int no_of_times_notout;
int total_runs;
double bat_avg;

CricketPlayer(String name, int no_of_innings, int no_of_times_notout,
        int total_runs) {
  this.name = name;
  this.no_of_innings = no_of_innings;
  this.no_of_times_notout = no_of_times_notout;
  this.total_runs = total_runs;
  this.bat_avg = 0;
}

static double avg(CricketPlayer p) {
  int outs = p.no_of_innings - p.no_of_times_notout;

  if (outs <= 0) {
    throw new ArithmeticException("Cannot calculate average");
  }

  return (double) p.total_runs / outs;
}

static void sortPlayer(CricketPlayer players[]) {
  for (int i = 0; i < players.length - 1; i++) {
    for (int j = i + 1; j < players.length; j++) {
```

```
            if (players[i].bat_avg > players[j].bat_avg) {
                CricketPlayer temp = players[i];
                players[i] = players[j];
                players[j] = temp;
            }
          }
        }
    }

    void display() {
        System.out.println("Name           : " + name);
        System.out.println("No. of Innings   : " + no_of_innings);
        System.out.println("Not Out        : " + no_of_times_notout);
        System.out.println("Total Runs      : " + total_runs);
        System.out.println("Batting Average   : " + bat_avg);
        System.out.println("----------------------------------------");
    }
}

public class CricketPlayerDemo {
    public static void main(String args[]) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of players: ");
        int n = sc.nextInt();
        sc.nextLine();
```

```
CricketPlayer players[] = new CricketPlayer[n];

for (int i = 0; i < n; i++) {

  System.out.println("\nEnter details of Player " + (i + 1));

  System.out.print("Enter Player Name: ");
  String name = sc.nextLine();

  System.out.print("Enter Number of Innings: ");
  int innings = sc.nextInt();

  System.out.print("Enter Number of Times Not Out: ");
  int notout = sc.nextInt();

  System.out.print("Enter Total Runs: ");
  int runs = sc.nextInt();
  sc.nextLine();

  players[i] = new CricketPlayer(name, innings, notout, runs);

  try {
    players[i].bat_avg = CricketPlayer.avg(players[i]);
  } catch (ArithmeticException e) {
    players[i].bat_avg = 0;
    System.out.println("Exception: " + e.getMessage());
  }
```

```
        }

        CricketPlayer.sortPlayer(players);

        System.out.println("\nPlayers Sorted According to Batting Average");
        System.out.println("=========================================");

        for (int i = 0; i < n; i++) {
            players[i].display();
        }

        sc.close();
    }
}
```

OR

2) Write a package for String operation which has two classes Con and Comp. Con class concatenates two strings, and comp class compares two strings. Also display proper message on execution.

Step 1- mkdir -p JavaProject/StringOperation

Step 2- cd JavaProject

Step 3- create a package

vim StringOperation/Con.java

```
package stringoperation;
 public class Con {
 public void concatenate(String str1, String str2)
 {
 String result = str1 + str2;
```

```
System.out.println("Concatenated String: " + result);
}
}
```

Step 4- vim StringOperation/Comp.java

```
package stringoperation;
public class Comp {
public void compare(String str1, String str2)
{
if (str1.equals(str2))
{
System.out.println("Both strings are equal.");
}
else
{
System.out.println("Both strings are not equal.");
}
}
}
```

Step 5: Main Program

vim StringOperationDemo.java

```
import java.util.Scanner;
import stringoperation.Con;
import stringoperation.Comp;
public class StringOperationDemo {
public static void main(String args[])
{
Scanner sc = new Scanner(System.in);
System.out.print("Enter first string: ");
```